



**UNIVERSITI TEKNOLOGI MARA**  
**FINAL EXAMINATION**

**COURSE : DISCRETE STRUCTURES**  
**COURSE CODE : CSC510**  
**EXAMINATION : JULY 2025**  
**TIME : 3 HOURS**

## INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (15 Questions)  
PART B (4 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
  - i) the Question Paper
  - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO**

*This examination paper consists of 8 printed pages*

**PART A (30 marks)**

1. Which of the following compound propositions is a tautology?
  - A.  $(P \rightarrow Q) \wedge (Q \rightarrow P)$
  - B.  $(P \vee Q) \rightarrow (P \wedge Q)$
  - C.  $(P \wedge \neg P) \rightarrow Q$
  - D.  $P \leftrightarrow \neg P$
  
2. Which of the following compound propositions is logically equivalent to an implication  $P \rightarrow Q$ ?
  - A.  $\neg P \vee Q$
  - B.  $\neg Q \rightarrow P$
  - C.  $\neg P \wedge Q$
  - D.  $P \leftrightarrow Q$
  
3. Which of the following is the contrapositive of the proposition  $(P \vee \neg Q) \rightarrow R$ ?
  - A.  $\neg R \rightarrow (\neg P \wedge Q)$
  - B.  $R \rightarrow (P \vee \neg Q)$
  - C.  $\neg R \rightarrow (\neg P \vee Q)$
  - D.  $\neg R \rightarrow (\neg P \wedge \neg Q)$
  
4. Given the propositions:  
P: "The system is secure"  
Q: "The system is accessible"  
Which of the following best represents the statement:  
**"If the system is not secure, then it must not be accessible"**?
  - A.  $\neg P \rightarrow \neg Q$
  - B.  $\neg Q \rightarrow \neg P$
  - C.  $P \rightarrow Q$
  - D.  $P \vee Q$
  
5. Find the contrapositive of the following statement:  
**"If it rains then the ground gets wet."**
  - A. If it does not rain, then the ground does not get wet.
  - B. If the ground does not get wet, then it does not rain.
  - C. If it rains, then the ground does not get wet.
  - D. If the ground gets wet, then it rains.

6. Find the converse of the following statement:  
"If the machine is on then the light is blinking."
- A. If the light is blinking, then the machine is on.
  - B. If the light is not blinking, then the machine is not on.
  - C. If the machine is on, then the light is not blinking.
  - D. If the machine is not on, then the light is not blinking.
7. Which of the following is logically equivalent to  $\neg \forall x P(x)$ .
- A.  $\forall x \neg P(x)$
  - B.  $\exists x \neg P(x)$
  - C.  $\neg \exists x \neg P(x)$
  - D.  $\exists x P(x)$
8. What is the **negation** of this predicate logic statement?  
 $\forall x (\text{Student}(x) \rightarrow \text{Studies}(x))$
- A.  $\forall x (\text{Student}(x) \wedge \neg \text{Studies}(x))$
  - B.  $\exists x (\text{Student}(x) \wedge \neg \text{Studies}(x))$
  - C.  $\exists x (\neg \text{Student}(x) \vee \text{Studies}(x))$
  - D.  $\forall x (\neg \text{Student}(x) \vee \text{Studies}(x))$
9. Which of the following is **NOT** a valid property of sets?
- A.  $A \cup B = B \cup A$
  - B.  $A \cap B = B \cap A$
  - C.  $A - B = B - A$
  - D.  $\emptyset \subseteq A$  for any set A
10. If U is the universal set and A is a subset of U, what is the **complement of A**?
- A.  $\{x \mid x \in A\}$
  - B.  $\{x \mid x \notin U\}$
  - C.  $\{x \mid x \in U \text{ and } x \notin A\}$
  - D.  $\{x \mid x \in A \text{ and } x \notin U\}$
11. If a relation R on a set A is reflexive, which of the following must be true?
- A.  $(a,b) \in R$  implies  $(b,a) \in R$
  - B.  $(a,a) \in R$  for all  $a \in A$
  - C.  $(a,b) \in R$  and  $(b,c) \in R$  implies  $(a,c) \in R$
  - D. R must be an equivalence relation

12. If a relation  $R$  on a set  $A$  satisfies  $(a,b) \in R \Rightarrow (b,a) \notin R$ , which of the following is true?
- A.  $R$  is symmetric
  - B.  $R$  is anti-symmetric
  - C.  $R$  is transitive
  - D.  $R$  is an equivalence relation
13. Which of the following strings cannot be generated by grammar  $(G)$  where  $S \rightarrow xyS \mid \varepsilon$ ?
- A.  $\varepsilon$
  - B.  $xyxyxy$
  - C.  $xxyy$
  - D.  $xy$
14. Which symbols appear in the final generated string of a grammar?
- A. Only non-terminal symbols
  - B. Only terminal symbols
  - C. Both terminal and non-terminal symbols
  - D. Neither terminal nor non-terminal symbols
15. Which of the following grammars belongs to Type 1 (Context-Sensitive Grammar)?
- A.  $A \rightarrow aB \mid \varepsilon$
  - B.  $AB \rightarrow BA$
  - C.  $A \rightarrow aBcD$
  - D.  $AB \rightarrow Aab$

**PART B (70 Marks)****QUESTION 1**

- a) Study the following statements about the process life cycle in a CPU:

If the process is running, then it has the CPU. If the process has a CPU and is not waiting, then the scheduler is active. The process is running and not in the waiting state.

Use the following representation to find the conclusion.

p: The process is terminated  
q: The process has the CPU  
r: The process is not in the waiting state  
s: The process is running  
t: The scheduler is active

- i) Write the premises derived from the above statement.

(3 marks)

- ii) Find a possible conclusion by using suitable rules of inference from your answer above.

(7 marks)

- b) By using indirect proof method, proof this statement:

If a number is divisible by 3, then it is divisible by 9.

(5 marks)

**QUESTION 2**

- a) In a universe of integer numbers, consider the following predicates:

$P(x)$ : x is a prime number

$E(x)$ : x is an even number

$D(x,y)$ : x is divisible by y

$O(x)$ : x is an odd number

- i) Express the statement in predicate:  
"All prime numbers greater than 2 are odd."

(2 marks)

- ii) Express the statement in predicate:  
"Some even numbers are divisible by 3."

(2 marks)



- iii) Given the statement  $\forall x (O(x) \rightarrow \neg E(x))$ , translate it into English. (2 marks)
- iv) If  $E(x)$  is true and  $P(x)$  is also true for some integer  $x$ , what might  $x$  be? (2 marks)
- b) Show an example of sets where we can get  $A \cap B$  and  $A \cap C$  which satisfy these conditions:
- $A \cap B \neq \emptyset$  and  $A \cap C \neq \emptyset$
- $B \neq C$
- $A \cap B \neq A \cap C$
- (7 marks)

### QUESTION 3

- a) A vending machine dispenses snacks based on a sequence of coin inputs. The system follows the grammar  $G = (V, S, T, P)$  where  $V: \{S, C, D, a, b\}$ , terminal symbols ( $T$ ):  $\{a, b\}$ , starting symbol:  $S$  and productions ( $P$ ) =  $\{S \rightarrow CD \mid C, C \rightarrow aC \mid a, D \rightarrow bD \mid b\}$ .
- Considering 'a' represents inserting a coin and 'b' represents selecting a snack. The machine dispenses a snack when at least one 'a' and one 'b' have been provided in sequence.
- i) Identify the type of grammar represented by the above production rules. Justify your answer. (2 marks)
- ii) Based on the production rules, generate a string that allows the vending machine to dispense snacks twice. (2 marks)
- iii) Determine whether the vending machine can dispense a snack if two coins are inserted before selecting a snack. Justify your answer using a derivation tree. (4 marks)
- b) A deterministic finite automaton (DFA) is widely used in real-world applications, especially in software and machine operations.
- i) Design a DFA-based application that verifies and accepts a secret password based on the following rules:
- Input =  $\{a, b, c, d\}$

The input must start with either "baca" or "abad"

The input must end with "abc"

Examples of accepted and rejected strings are shown below:

Accepted:

bacaabc, bacabbbbaababc, bacaaaababc, bacababc, abadabc, abaddddaabc, abadaaabbdabc

Rejected:

abcaabaca, bdbacaabc, abdabd, baccabc, a, ab, c, d, abcd

(8 marks)

- ii) Based on your design in (i), complete the state transition table for the DFA.

(4 marks)

#### QUESTION 4

- a) Find the pre-condition and post-condition for the statement below. You are required to provide the answer using the Hoare Triple table.

i)  $\{b=3\} \ a := b + 4; \ b := a - 2; \ \{ ? \}$

(1.5 marks)

ii)  $\{ ? \} \ y := 2x+5; \ \{y>10\}$

(1.5 marks)

iii)  $\{ ? \} \ x := a + b; \ y := x * 2; \ z := y - b; \ \{z = 2a + b\}$

(3 marks)

- b) Prove the validity of the following decision segment that can guarantee the post-condition  $\{z > -1\}$  to be true using Hoare Triple approach.

```

if (x > 0) then {
    y := x * 2;
    z := y + 1; }
else {
    y := -x;
    z := y - 1; }
end if

// post-condition: {z > -1}

```

(6 marks)

c) Answer all questions below regarding the repetition program structure.

i) Formulate the loop invariant for the code below.

```
int y = 1;
int k = 0;

while (k != 10) {
    y = y * 2;
    k = k + 1;
}
```

(2 marks)

ii) Prove the loop invariant created from i) using the Hoare Triple approach.

(4 marks)

iii) Identify the strongest post-condition for the program shown below using the Hoare Triple approach. If the pre-condition of the while loop is given as  $\{m \geq 4\}$  and the loop invariant is  $\{x = 5^j\}$ .

```
int x = 1, j = 0;

while (j < m) {
    x = x * 3;
    j = j + 1;
}
```

(2 marks)

**END OF QUESTION PAPER**